Etanercept (Injection)

- Type: Biologic Disease-Modifying Antirheumatic Drug (bDMARD)
- Mechanism of Action: Tumor Necrosis Factor (TNF) inhibitor; reduces inflammation by blocking TNF-alpha

Side Effects: Increased risk of infections, injection site reactions, headache, nausea, immune suppression

Methotrexate (Tablet)

- Type: Conventional Disease-Modifying Antirheumatic Drug (cDMARD)
- Mechanism of Action: Inhibits dihydrofolate reductase, affecting DNA synthesis and reducing immune system activity

Side Effects: Nausea, liver toxicity, bone marrow suppression, fatigue, mouth ulcers.

Combination Use

- Often prescribed together in autoimmune diseases like RA to enhance efficacy and reduce antibody formation against etanercept

Rheumatoid Arthritis (RA) – Brief Overview

1. Etiology (Causes)

RA is an autoimmune disorder where the immune system mistakenly attacks the synovium (lining of the joints).

  - Genetic factors: HLA-DR4, HLA-DR1 (associated with increased susceptibility).
  - Environmental triggers: Infections, smoking, gut microbiota imbalance.
  - Hormonal factors: More common in females, suggesting estrogen involvement.

2. Pathophysiology

  - Trigger: Autoimmune response activates T-cells, B-cells, and macrophages.
  - Cytokine Release: TNF-α, IL-1, IL-6 promote inflammation.
  - Synovial Inflammation: Leads to synovial hyperplasia (pannus formation).
  - Joint Destruction: Bone erosion (osteoclast activation), cartilage damage.

- Systemic Effects: Extra-articular manifestations (lungs, heart, eyes, vasculitis).

3. Risk Factors

- Genetic: HLA-DR4, HLA-DR1.
- Gender: Women > Men (3:1).
- Age: Common onset between 30-50 years.
- Smoking: Strongest environmental risk factor.
- Infections: Epstein-Barr virus (EBV), periodontal disease.
- Obesity & Diet: May influence inflammation levels

Naproxen Sodium BID vs QD

- BID (Twice Daily): Preferred for chronic inflammatory conditions (e.g., rheumatoid arthritis, osteoarthritis) due to sustained pain relief and anti-inflammatory effects.
- QD (Once Daily): Less common but may be used for mild pain or short-term relief. Some extended-release formulations allow QD dosing

Febuxostat

- Class: Xanthine oxidase inhibitor (like allopurinol).
- Indication: Chronic gout management (lowers uric acid levels)

Right MTP Joint Involvement

- MTP (Metatarsophalangeal) Joint Involvement: Often affected in gout, rheumatoid arthritis, osteoarthritis.
- Common Causes:
- Gout: Sudden onset, severe pain, redness, swelling (often first attack site).
- RA: Symmetric joint involvement, morning stiffness.
- Osteoarthritis: Chronic pain, worsens with activity

Acute Gout

- Definition: Sudden, severe inflammatory arthritis due to uric acid crystal deposition in joints.

- Symptoms:
- Rapid-onset severe pain, redness, warmth, and swelling.
- Common site: 1st MTP joint (podagra), but can affect other joints.
- Peaks within 24 hours, lasts days to weeks.
- Triggers: Alcohol, purine-rich foods (red meat, seafood), dehydration, medications (diuretics, aspirin).
- Treatment:
- First-line: NSAIDs (e.g., naproxen, indomethacin).
- Alternative: Colchicine (within 24 hours of attack), corticosteroids.

2. Chronic Gout

- Definition: Recurrent gout attacks leading to chronic inflammation, joint damage, and tophi formation.
- Symptoms:
- Frequent attacks, persistent joint pain, tophi (deposits of urate crystals).
- Can cause joint deformity and kidney stones.
- Risk Factors: Uncontrolled hyperuricemia (>6.8 mg/dL), renal dysfunction, obesity, alcohol use.
- Treatment:
- Urate-lowering therapy (ULT):
- First-line: Allopurinol or febuxostat (xanthine oxidase inhibitors).
- Alternative: Probenecid (uricosuric agent).
- Colchicine/NSAIDs for flare prophylaxis during initial ULT therapy

Etiology (Causes)

Gout is caused by hyperuricemia (serum uric acid >6.8 mg/dL), leading to urate crystal deposition in joints.

- Primary Gout (90%) – Due to genetic/metabolic defects in uric acid metabolism.

- Secondary Gout (10%) – Caused by underlying conditions or medications.

Causes of Hyperuricemia:

- Increased Uric Acid Production:
- High-purine diet (red meat, seafood, alcohol).
- Genetic disorders (Lesch-Nyhan syndrome, PRPP synthase overactivity).
- High cell turnover (e.g., hemolysis, chemotherapy, psoriasis).
- Decreased Uric Acid Excretion (More Common Cause):
- Renal impairment (CKD) – Reduced urate clearance.
- Diuretics (thiazides, loop diuretics) – Promote urate retention.
- Aspirin (low dose) – Competes with uric acid excretion.
- Metabolic disorders (obesity, insulin resistance)

Pathophysiology

1. Hyperuricemia: Uric acid accumulates in the blood.
2. Crystal Deposition: Monosodium urate (MSU) crystals deposit in the synovium, triggering immune response.
3. Inflammatory Cascade:

- Neutrophils and macrophages try to engulf MSU crystals.
- Cytokines (IL-1, TNF-α) and chemokines amplify inflammation.
- Leads to intense pain, redness, swelling, and joint destruction if chronic.

4. Tophi Formation (Chronic Gout): Hard, urate deposits develop in joints, tendons, and soft tissues

Risk Factors

- Non-Modifiable:
    - Male sex (higher uric acid levels).
    - Age >40 years.

    - Genetic predisposition (e.g., HLA-B58).
- Modifiable:
    - Diet: High purine intake (meat, seafood, alcohol, fructose).
    - Obesity & Metabolic Syndrome: Insulin resistance reduces urate clearance.
    - Chronic Kidney Disease (CKD): Impaired urate excretion.
    - Medications: Thiazides, loop diuretics, low-dose aspirin, cyclosporine.
    - Dehydration & Alcohol:

Aerobic respiration provides long-term energy for sustained activities and prevents lactic acid buildup.

Anaerobic respiration kicks in during intense exercise when oxygen is limited but causes muscle fatigue due to lactic acid accumulation